

Project Proposal

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Programme: MSc Applied Data Science

Course: COMP6830 – Data Science Capstone II

Project Title:

Scalable Music Genre Classification and Audio Analytics Using the Free Music Archive (FMA-Large) Dataset

1. Domain Background

The exponential growth of digital music platforms has resulted in large-scale repositories containing hundreds of thousands of audio tracks. Effective organisation, retrieval, recommendation, and archival of such repositories require reliable automated classification systems. Manual genre labelling is subjective, inconsistent across curators, and not scalable for collections exceeding tens of thousands of tracks.

Music Information Retrieval (MIR) applies signal processing, statistics, and machine learning techniques to extract structured information from raw audio signals. One of the most fundamental MIR tasks is **music genre classification**, which underpins recommendation engines, similarity search, playlist generation, and metadata enrichment. Despite its popularity, genre classification remains challenging due to overlapping genre boundaries, hierarchical taxonomies, stylistic evolution, and class imbalance.

The Free Music Archive (FMA) dataset was introduced to address limitations of small or proprietary MIR datasets by providing a large-scale, open, and research-oriented benchmark (Defferrard et al., 2017). The dataset contains 106,574 tracks, 161 hierarchically organised genres, and approximately 917GB of Creative Commons–licensed audio. This scale introduces realistic **big-data challenges**, including storage architecture, data pipelines, computational efficiency, and experiment reproducibility.

This project addresses the need for a scalable, reproducible data science pipeline capable of handling large audio repositories while delivering interpretable genre classification models and analytical insights. The work aligns with applied data science principles by integrating data engineering, modelling, evaluation, and deployment considerations within a unified framework.

2. Objectives of the Project

2.1 Business Objectives

Upon completion, the project aims to:

- Develop a scalable, automated genre classification system suitable for large music repositories.

- Demonstrate how structured metadata and engineered audio features can improve digital music organisation and retrieval.
- Provide analytical insights (e.g., genre distributions, feature importance patterns) to support music discovery, archival management, and recommendation systems.
- Establish a reproducible data pipeline that can be extended to other multimedia classification domains.

2.2 Research Objectives

- Evaluate the effectiveness of engineered audio features (e.g., MFCCs, spectral descriptors) in large-scale, multi-class genre classification.
 - Compare the performance, scalability, and interpretability of multiple machine learning techniques under big-data conditions.
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3. Description of Data

3.1 Data Domain

The data consists of digital music audio files and structured metadata from the Free Music Archive (FMA), an open-access repository designed for music analysis research.

3.2 Data Source and Repository Structure

The dataset will be obtained from the official Free Music Archive (FMA) GitHub repository and associated metadata archives. All metadata and engineered features for the complete dataset are distributed in **fma_metadata.zip (~342 MiB)**. These structured files can be accessed using Python (e.g., pandas) or other analytical tools.

The metadata archive contains the following core files:

- **tracks.csv** – Track-level metadata for all 106,574 tracks, including track ID, title, artist, genre labels, tags, and play counts.
- **genres.csv** – 163 genres with parent relationships, enabling hierarchical genre inference and aggregation.
- **features.csv** – Pre-computed audio features extracted using *librosa*, including spectral, timbral, and rhythmic descriptors.
- **echonest.csv** – High-level audio descriptors (e.g., energy, danceability) for a subset of 13,129 tracks.

In addition to metadata, the repository provides multiple MP3-encoded audio archives of varying sizes:

- **fma_small.zip** – 8,000 tracks (30s), 8 balanced genres (~7.2 GiB)
- **fma_medium.zip** – 25,000 tracks (30s), 16 unbalanced genres (~22 GiB)
- **fma_large.zip** – 106,574 tracks (30s), 161 unbalanced genres (~93 GiB)
- **fma_full.zip** – 106,574 full-length tracks, 161 unbalanced genres (~879 GiB)

3.3 Dataset Selection and Rationale

This project will utilise **FMA-Large (~93 GiB)** as the primary working dataset.

Rationale for selecting FMA-Large: - It contains the complete set of 106,574 tracks while remaining computationally feasible for modelling experiments. - The 161 unbalanced genres reflect real-world distribution patterns, enabling realistic evaluation of multi-class classification under class imbalance. - The dataset size satisfies big-data expectations while maintaining manageable storage and processing requirements compared to FMA-Full (~879 GiB).

Where computational resources permit, limited experiments may explore FMA-Full for scalability discussion; however, FMA-Large will serve as the core dataset for modelling and evaluation.

3.4 Metadata and Feature Usage

- tracks.csv provides primary genre labels and identifiers.
- genres.csv enables hierarchical aggregation and multi-level analysis.
- features.csv contains engineered descriptors including MFCCs, spectral centroid, bandwidth, roll-off, and zero-crossing rate.
- Metadata-provided split indicators (training/validation/test) will be preserved to ensure reproducibility and benchmarking consistency.

3.5 Storage and Data Management

Given the dataset size (~93GB audio), a two-layer storage architecture will be implemented.

File Storage Layer

- Audio files will remain stored in structured directories as distributed.
- Raw metadata CSV files will be version-controlled to maintain reproducibility.

Database Layer (MongoDB)

A document-oriented database (MongoDB) will be used to store unified track records containing:

- track_id
- audio_path
- split (train/validation/test)
- genre and hierarchical labels
- metadata attributes
- selected feature vectors

Rationale: - Supports semi-structured data (hierarchical genres, tag arrays). - Enables efficient retrieval of partitioned datasets. - Facilitates experiment tracking and reproducibility.

3.6 Data Pipeline Overview

The end-to-end data pipeline will consist of:

1. Download and checksum validation of FMA archives.

2. Extraction of audio and metadata files.
3. Parsing and joining metadata tables via track_id.
4. Loading unified records into MongoDB.
5. Preserving predefined train/validation/test splits.
6. Supplying partitioned data to modelling scripts.

This pipeline ensures transparency in how raw data becomes modelling-ready inputs.

4. Data Science Objectives

The project follows the CRISP-DM framework.

4.1 Business Understanding

Translate the domain goal (automated genre classification for large repositories) into measurable predictive modelling tasks.

4.2 Data Understanding and Preparation

- Exploratory analysis of genre distributions and class imbalance.
- Feature normalization and scaling.
- Verification and preservation of predefined dataset splits.
- Optional hierarchical aggregation of genres.

4.3 Model Creation

Four modelling techniques will be developed and compared:

1. Logistic Regression (baseline linear classifier).
2. Support Vector Machines (non-linear modelling).
3. Random Forest Classifier (ensemble modelling and feature importance).
4. Convolutional Neural Networks using spectrogram representations.

This staged modelling approach allows evaluation of increasing model complexity under identical data conditions.

4.4 Evaluation

Models will be evaluated using: - Accuracy - Precision, Recall, F1-score - Confusion matrices - Cross-model comparative analysis

4.5 Deployment and Application

- Reproducible training and evaluation scripts.
 - Batch prediction capability for unseen tracks.
 - Demonstration of integration into music discovery or archival systems.
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5. Key Deliverables for the Project

Aligned with the course outline and objectives:

Technical Artefacts

- Implemented data ingestion and MongoDB-backed storage pipeline.
- Cleaned and partitioned analytical dataset.
- Four trained classification models.
- Comparative performance report.

Analytical Outputs

- Genre distribution visualisations.
- Feature importance analyses.
- Confusion matrix evaluations.

Applied Outputs

- Reproducible modelling pipeline.
- Demonstration prediction interface or script.

These deliverables directly support the business objectives of scalable classification and improved digital music organisation.

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